# Evaluation: study/main/cifar10-dit-s106/evaluations/test-nfe32

## Purpose and interpretation

This evaluation measures a fixed checkpoint and fixed noise population. KID subset SD describes metric subsampling variability; it is not a confidence interval across trained seeds. The first 64 images are shown without selection.

## Recorded evidence


```
{
  "dataset": "cifar10",
  "family": "dit",
  "training_seed": 106,
  "split": "test",
  "intervention": "none",
  "generated_count": 5000,
  "real_count": 10000,
  "sampling_seed": 20261016,
  "metric_seed": 20261016,
  "nfe": 32,
  "kid_subsets": 100,
  "kid_subset_size": 1000,
  "checkpoint_sha256":
"c05ea7f6e8db40d1fee8137f96251a02fd2c1ee199d91e2214aa46a17c51cc5e",
  "data_sha256":
"51ca348f81e4c280d10942cf2fa0d4dd1ce688a16b0088283069b54bbaaf6f97",
  "source_revision": "663de2267e5c66dd3d56cae3c538359519f635f4",
  "primary_eligible": true,
  "device": "cuda",
  "blas_threads": 4,
  "status": "completed",
  "sampling": {
    "count": 5000,
    "seed": 20261016,
    "nfe": 32,
    "batch_size": 32,
    "seconds": 48.83142051199684,
    "seconds_per_image": 0.009766284102399369,
    "precision": "float32",
    "timing_excludes": "model load and warmup",
    "clipped_fraction": 0.015461849048733711,
    "peak_memory_bytes": 58955776
  },
  "metrics": {
    "frechet_inception_distance": 106.24901616082059,
    "kernel_inception_distance_mean": 0.09459355115890503,
    "kernel_inception_distance_std": 0.0018841878878409941
  },
  "validation_velocity_mse": 0.20871861185878515,
  "metric_seconds": 20.633361776999664,
```

```
  "id": "study/main/cifar10-dit-s106/evaluations/test-nfe32",
  "artifact": "runs/study/main/cifar10-dit-s106/evaluations/test-nfe32/
evaluation.json"
}
```


## Fixed sample grid

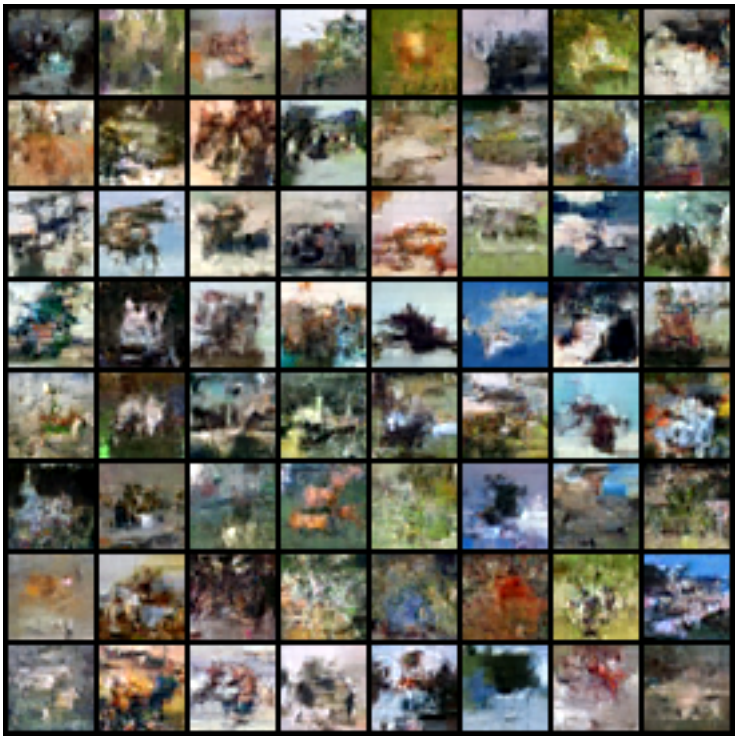


Figure 1: First 64 generated samples

## Method choices and learning

The shared optimizer, data policy, architecture, and stopping rule were chosen before final comparison and are described in the technical paper. The 3e-4 learning rate is a conservative AdamW starting point, not a tuned optimum. Microbatches bound laptop memory; four internal updates bound recurrent cost; EMA smooths noisy parameter updates. These choices trade study breadth for a fixed local compute budget. This record supports only the measurements above; causes of quality differences require the trained controls and uncertainty analysis. Full trajectories, populations, RNG states, and logs remain in the referenced artifact bundle. Sources: protocol, TorchCFM, AKOrN.